

PSOS (Principles of Secured Operating Systems) 2024/2025 - Coursework Marking Rubric

Criteria	First		Second Class-First Division	Second Class – Second Division	Third	Refer/Fail		Weighting 100
	100 – 85%	84 – 70%	69 – 60%	59 – 50%	49 – 40%	39 – 25%	24 – 0%	
	A+	A	B+/B	C+/C	D+/D	Refer/Fail	Refer/Fail	
Section 1 A description of File Encryption	Excellent Description! It provides an excellent overview of File Encryption and almost all essential roles and features of File Encryption	A very good description that provides a very good overview of File Encryption and the majority of essential roles and features of File Encryption.	The description gives a reasonable background of the File Encryption and the majority of essential roles and features of File Encryption.	The description provides a basic background of the File Encryption, but is vague about the majority of essential roles and features of File Encryption	Satisfactory: Provides an acceptable overview of File Encryption and some of its roles and features.	The description of File Encryption was inadequately addressed, poor quality content, incomplete /irrelevant.	The description of the file Encryption fails to meet the requirements/Missing	5 Marks
Section 2 A list of functional and non-functional requirements and security features of File Encryption	An outstanding list that clearly identifies and categorizes functional and non-functional requirements, along with detailed security features. Each requirement is specific, relevant, and thoroughly explained, reflecting deep understanding and excellent organization	A strong list that effectively covers functional and non-functional requirements and security features. Minor areas could be expanded for deeper clarity or more examples, but overall, the work shows solid comprehension and logical structure.	A good list that addresses functional and non-functional requirements and mentions security features, though some elements may lack detail or clarity. The organization is sufficient but could be improved with more precise definitions or examples.	A basic list that identifies the main categories but provides limited detail or uneven coverage. Some requirements or security aspects may be unclear or missing. Better organization and elaboration are needed to fully meet expectations.	The list is minimal and omits key functional, non-functional, or security requirements. The structure is weak, and important details are missing. Significant expansion and clarification are necessary to improve the submission.	The list fails to meet expectations, with vague or irrelevant points. Most required elements (functional, non-functional, and security features) are missing or poorly explained. A major rewrite is required to align with the task.	The list is missing, off-topic, or does not address functional, non-functional requirements, or security features. It shows little to no understanding of the task and needs a complete overhaul.	10 Marks

Section 3 Design of your software /patch that includes communications with the selected OS	Outstanding body of work! It thoroughly and comprehensively addresses all required sections with excellent detail and clarity. Each subsection is well-organized and demonstrates a deep understanding of the topic. The analysis is insightful, and the use of examples or case studies enhances the discussion. The comparative analyses and case studies are highly relevant and well-articulated.	A very good body of work that covers all required sections with clear and well-organized content. There is a good level of detail and understanding demonstrated in each subsection. The analysis is strong, but there might be minor areas where more depth or examples could enhance the discussion. Overall, it is a high-quality submission.	A good body of work that addresses the required sections adequately. The content demonstrates understanding but lacks depth or critical analysis in some subsections. Examples of case studies may be present but are not well-integrated or detailed. Improved organization and elaboration in some areas would strengthen the submission.	The body of the coursework provides a basic overview of the required sections but lacks depth, clarity, or coherence in several areas. Subsections might be underdeveloped or missing key details. Analysis is minimal, and examples or case studies are either absent or insufficiently discussed. More structure and critical engagement are needed.	The body of the coursework is poorly developed and misses key components or sections. Content is vague or irrelevant in some areas, and there is little evidence of understanding or critical analysis. Examples of case studies are either irrelevant or absent. Significant improvements in structure, content, and detail are required.	The body of the coursework fails to meet expectations, with many required sections missing or inadequately addressed. Content lacks coherence, and there is minimal evidence of understanding or analysis. Examples of case studies are either incorrect or entirely missing. A substantial rewrite is necessary to meet the criteria.	The body of the coursework is either missing or entirely irrelevant. None of the required sections are adequately addressed, and there is no evidence of understanding or effort to meet the rubric criteria. A complete overhaul is required to align with expectations.	10 Marks
Section 4 Implementation of your File Encryption, including annotated C code	An outstanding implementation with fully functional and well-structured C code. The code is thoroughly annotated with clear, insightful comments explaining the logic and flow. Demonstrates a deep understanding of file encryption concepts and excellent coding practices.	A strong implementation with working C code and helpful annotations. Most parts of the code are clearly explained, though a few areas could benefit from slightly more detailed comments. Shows strong understanding and good coding standards.	A good implementation where the C code functions correctly but may lack depth or consistency in annotations. Some parts are well explained, while others are brief or unclear. The coding structure is generally sound but could be refined.	A basic implementation that attempts file encryption but has limited or inconsistent commenting. The code may work partially or require minor corrections. Annotations are present but need more clarity to explain the full logic and purpose. .	The implementation is incomplete, has major errors, or is missing key parts of the file encryption process. Annotations are sparse, unclear, or missing. Significant revisions are needed in both code functionality and clarity of comments.	The implementation does not meet expectations. The C code is largely non-functional, disorganized, or irrelevant to file encryption. Very few or no meaningful comments are provided. A full rewrite of the code and proper annotation are required.	The implementation is missing, off-topic, or shows no understanding of C programming or file encryption. There is no useful code or annotation present. A complete and thorough redevelopment is necessary.	15 Marks

Section 5 Testing plan for validating your software	An outstanding testing plan that clearly outlines detailed test cases, expected outcomes, and validation methods. The plan is thorough, systematic, and demonstrates a deep understanding of how to effectively validate the software's functionality, security, and reliability.	A strong testing plan that covers key test cases and expected outcomes. It shows good organization and a solid approach to validating the software, though a few minor areas could benefit from more detailed testing strategies or additional scenarios.	A good testing plan that identifies several test cases and describes expected results, though some tests may be missing or underdeveloped. The validation approach is clear but could be strengthened with more thorough coverage or better structure.	A basic testing plan that covers essential tests but lacks depth or breadth. Some key validation steps are missing or not clearly explained. The structure is understandable but needs more detail to fully ensure the software's reliability.	A limited or incomplete testing plan that addresses only a few basic test cases. The validation strategy is weak, and important aspects of the software may not be tested. Significant improvements are needed in both the range and explanation of test cases.	The testing plan does not meet expectations. Very few meaningful tests are described, and there is little or no clear validation strategy. Most important aspects of the software are left untested. A major revision is required.	The testing plan is missing, off-topic, or shows no understanding of software validation. There are no meaningful test cases or strategies provided. A complete and detailed testing plan needs to be developed from scratch.	10 Marks
Section 6 Description of integrating/adding the implemented component to the OS	An outstanding description that clearly and thoroughly explains the steps taken to integrate the implemented component into the operating system. It covers installation paths, commands used, permissions, and verification methods with precision, demonstrating a deep understanding of system-level integration.	A strong description that explains the integration process effectively, including most relevant steps like moving files, setting permissions, and verifying system-wide access. Minor areas could be expanded with slightly more detail or explanation for full clarity.	A good description that outlines the main steps for adding the component to the OS, but some parts (such as permissions, verification, or error handling) may be missing or not fully explained. The structure is acceptable but could benefit from a more complete walkthrough.	A basic description that touches on key points of integration but lacks depth or skips important steps. Some actions may be unclear or incomplete, and additional detail is needed to fully demonstrate how the component becomes available system-wide.	A limited or incomplete description that only partially explains the integration process. Important steps such as correct file placement, permissions, or testing system-wide availability are missing or unclear. Significant improvements are needed to meet expectations.	The description does not meet expectations. It is vague, missing essential integration steps, or incorrect in method. There is little evidence of understanding how to properly add a component to the operating system. A full rewrite with accurate technical detail is required.	The description is missing, off-topic, or shows no understanding of system integration. There are no meaningful steps or explanations provided. A complete and accurate description needs to be developed from the beginning.	5 Marks
Section 7 Integration testing plan for integrating your component	An outstanding integration testing plan that thoroughly outlines test cases, expected outcomes, and verification steps. It clearly demonstrates how	A strong integration testing plan that covers major test cases and system-level validations. The steps are mostly clear and logical, though a	A good integration testing plan that addresses the basic tests needed to verify system integration. Some tests are well explained, but	A basic integration testing plan that identifies a few tests related to adding the component into the system, but misses some important checks (like	A limited or incomplete integration testing plan that only partially covers how the component's integration is verified. Important	The integration testing plan does not meet expectations. It lacks most required tests, provides unclear steps, or shows weak	The integration testing plan is missing, irrelevant, or completely off-topic. It shows no understanding of how to verify that a component has been	5 Marks

nt into the system	the component's functionality is validated within the system environment, covering permissions, execution from any directory, error handling, and overall system behavior. Shows deep understanding and excellent organization.	few areas (such as testing unusual scenarios or additional system responses) could be expanded for more completeness.	others may lack depth or completeness. The overall structure is acceptable but could benefit from additional coverage or refinement.	permission errors or path issues). The plan demonstrates a general understanding but needs more detail and better organization.	system behaviors, error cases, or validation steps are missing or unclear. Significant development of the testing plan is needed.	understanding of system-wide behavior. Little to no practical validation is described. A major rewrite with full system testing is required.	correctly added to an operating system. A complete and detailed testing plan needs to be written from scratch.	
Section 8 Reporting the possible limitations, failures, and/or difficulties you experience in your work	An outstanding report that clearly and thoughtfully identifies possible limitations, failures, and difficulties encountered during the work. The discussion is honest, detailed, and reflective, showing deep insight into challenges faced and how they were addressed or could be improved.	A strong report that effectively discusses key limitations, failures, and difficulties. The explanations are mostly clear and well-organized, though a few areas could be expanded with deeper reflection or more specific examples for greater depth.	A good report that mentions important limitations, failures, and difficulties but may lack detailed explanations or critical analysis. The discussion is clear but could be improved by connecting challenges more strongly to the work's outcomes or lessons learned.	A basic report that acknowledges some limitations or difficulties but lacks depth, completeness, or clear reflection. Some key challenges may be missing or only briefly mentioned. Better organization and fuller explanation are needed to strengthen the section.	A limited or incomplete report that briefly mentions few challenges without meaningful detail or analysis. Important limitations, failures, or difficulties are missing or unclear. Significant improvement is needed in identifying and reflecting on work challenges.	The report does not meet expectations. Very few or no limitations, failures, or difficulties are reported. The discussion is vague, superficial, or poorly organized, showing little understanding of critical reflection. A full rewrite is necessary.	The report is missing, off-topic, or shows no effort to identify or reflect on limitations, failures, or difficulties experienced during the work. A complete, thoughtful report needs to be developed from the beginning.	5 Marks
Section 9 A conclusion section that includes recommendations for extending the	An outstanding conclusion that clearly summarizes the work conducted and thoughtfully provides insightful, practical recommendations for future extensions or improvements. The section shows deep	A strong conclusion that effectively summarizes the main work and presents good recommendations for extension. A few areas could be expanded with deeper reflection or more specific future ideas, but overall it	A good conclusion that touches on the key achievements and suggests reasonable extensions. Some recommendations may be general or briefly explained, and a stronger link between the current work and future	A basic conclusion that summarizes the work but gives limited or vague recommendations for extension. The section shows general understanding but lacks depth or specific, forward-looking ideas. More	A limited or incomplete conclusion that only briefly summarizes the work without meaningful or thoughtful recommendations. Important aspects of extending the work are missing or underdeveloped.	The conclusion does not meet expectations. It lacks a clear summary and gives little or no useful recommendations for future work. The content is vague, unfocused, or poorly organized. A full rewrite is	The conclusion is missing, off-topic, or fails to address both the summary of the work and the recommendations for extension. A complete, thoughtful conclusion section needs to be	5 Marks

conducted work	reflection, strong understanding of the project's broader potential, and excellent organization.	demonstrates solid comprehension and critical thinking.	ideas would improve the overall impact.	detailed reflection and future planning would strengthen it.	Significant improvements are needed to meet expectations.	necessary to properly close the report.	developed from scratch.	
Section 10 Description and reflections on improvements/revisions made. This section is critical as it highlights the important progress made in the current version of your coursework	An outstanding description and reflection that thoroughly explains the improvements and revisions made, clearly highlighting the important progress achieved. The reflection is deep, thoughtful, and shows excellent critical analysis of the changes and their impact on the quality of the work.	A strong description and reflection that effectively discusses the key improvements and revisions. Most changes are clearly explained and justified, though deeper reflection on how these changes strengthened the work would make the section even better.	A good description that covers the main revisions and some improvements, but the reflection on their significance is brief or general. The section would benefit from a more detailed discussion of how and why the changes improved the coursework.	A basic description that mentions some improvements or revisions but lacks depth and detailed reflection. The changes are noted but not fully analyzed in terms of how they contributed to progress. More critical thinking and organization are needed.	A limited or incomplete description that only briefly touches on revisions, without clearly explaining the nature or impact of the improvements. Significant development is needed to properly reflect on how the coursework was strengthened.	The section does not meet expectations. Few or no revisions are clearly described, and little to no reflection is provided on how improvements were made. The work shows little understanding of the importance of revising and improving. A major rewrite is required.	The section is missing, off-topic, or shows no effort to describe or reflect on revisions or improvements. There is no evidence of progress made. A complete, thoughtful rewrite is necessary to meet expectations.	15 Marks
Section 11 Includes only references (IEEE referencing Style) and	An outstanding references section that strictly follows IEEE referencing style with perfect formatting. All sources are relevant, properly cited, and consistently formatted.	A strong references section that correctly uses IEEE style with only minor formatting inconsistencies. Sources are appropriate and mostly well-cited, with just a few	A good references section that mostly follows IEEE style, though there are some noticeable formatting errors or inconsistencies. The sources are relevant, but greater care in formatting and	A basic references section that attempts to follow IEEE style but contains several formatting mistakes or inconsistencies. Some sources may be improperly cited or incomplete. More careful editing and	A limited references section that shows weak application of IEEE style, with frequent formatting issues or missing required details. The organization is unclear, and significant	The references section does not meet expectations. It either ignores IEEE formatting rules or includes very few correctly formatted citations. The sources may also be	The references section is missing, completely off-format, or irrelevant. There is no meaningful attempt to follow IEEE referencing style. A complete rewrite is needed,	5 Marks

Formattin g	Demonstrates excellent attention to detail and a strong understanding of academic standards.	small improvements needed for perfect consistency.	checking against the IEEE standard would strengthen it.	closer adherence to the IEEE format are needed.	improvements are needed to meet proper referencing standards.	inappropriate or poorly selected. A full rework is necessary.	following correct academic standards.	
Section 12 Includes only Reference and Formattin g	Demonstrates an outstanding understanding of multi-threaded and multi-process systems, clearly applying key concepts with precision and depth. Participation in seminar classes and exercises is excellent, with well over 50% attendance and active engagement.	Shows a strong understanding of multi-threaded and multi-process systems, with minor gaps or areas for deeper explanation. Seminar class participation is good, with attendance above 50% and consistent engagement in exercises.	Demonstrates a good understanding of multi-threaded and multi-process systems, though some explanations may lack depth or clarity. Seminar attendance is above 50%, and there is satisfactory participation in class exercises.	Shows a basic understanding of multi-threaded and multi-process systems but misses key details or has some misunderstandings. Attendance meets the 50% minimum, with basic participation in class activities and exercises.	Demonstrates limited understanding of multi-threaded and multi-process systems, with significant gaps or misconceptions. Attendance is just above 50%, but participation in exercises and discussions is minimal or passive.	Shows weak or superficial understanding of multi-threaded and multi-process systems, with many inaccuracies. Although attendance is technically above 50%, there is little evidence of active participation in class exercises.	Fails to demonstrate understanding of multi-threaded and multi-process systems. Attendance may barely meet the minimum, but participation is absent or irrelevant. A major improvement in both comprehension and engagement is required.	10 Marks